



Rock, Paper, Scissors, Lizard, Spock

A C Programming Mini-Project

INTRODUCTION

Beyond the Basics: A CS Adventure

Welcome to our mini-project presentation! We're diving into the classic Rock, Paper, Scissors game, but with a twist inspired by the popular "Big Bang Theory" adaptation: adding Lizard and Spock. This project demonstrates fundamental C programming concepts in a fun, interactive way.



C Programming

Reinforcing core language skills.



Game Logic

Implementing decision-making rules.



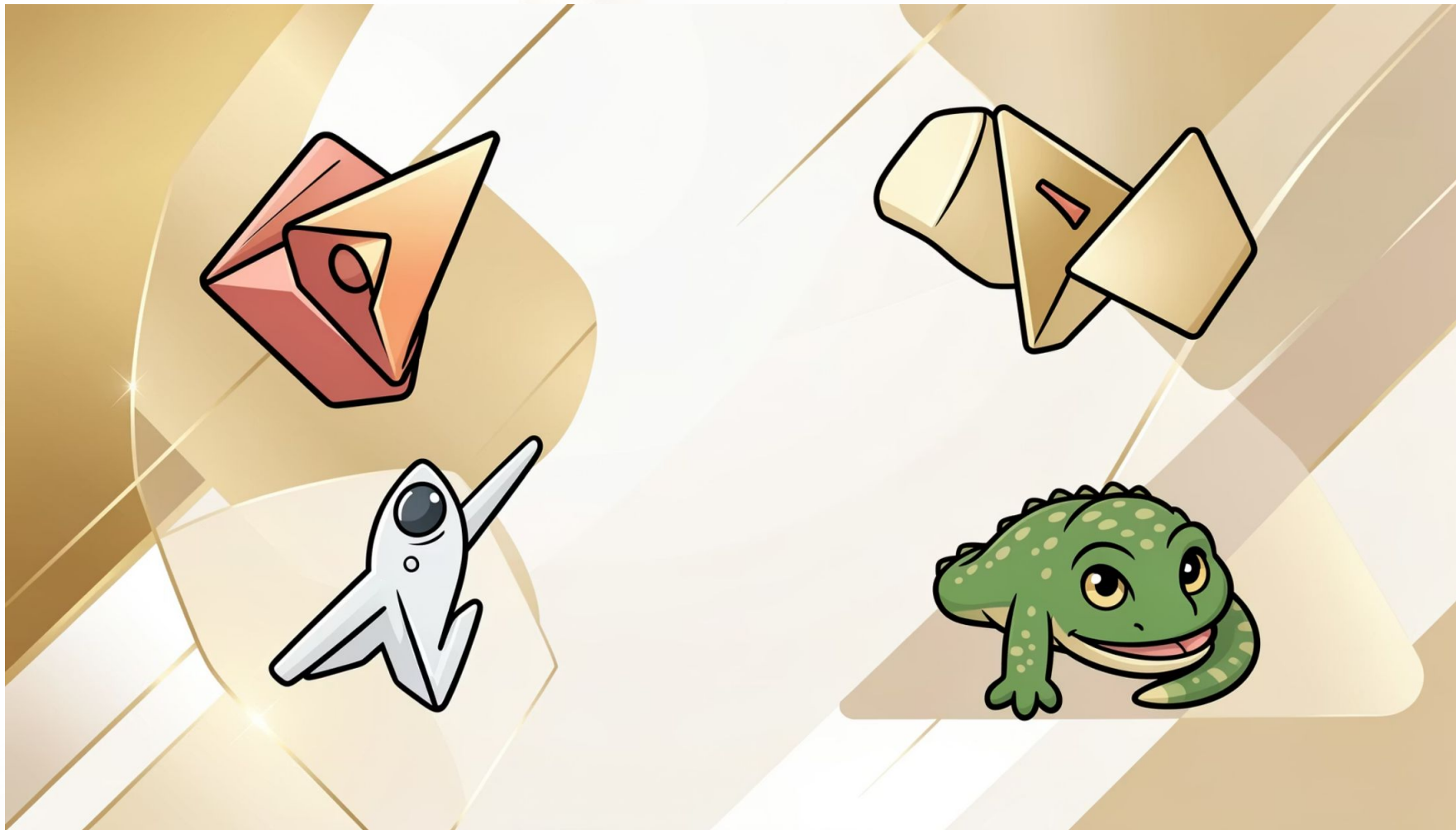
Problem Solving

Designing and debugging an interactive program.

THE
GAME

Understanding the Rules of Engagement

The expanded game adds two new elements, significantly increasing the strategic depth beyond the traditional three choices. It's not just luck; it's about anticipating your opponent!



Rock crushes Scissors, crushes Lizard

Paper covers Rock, disproves Spock

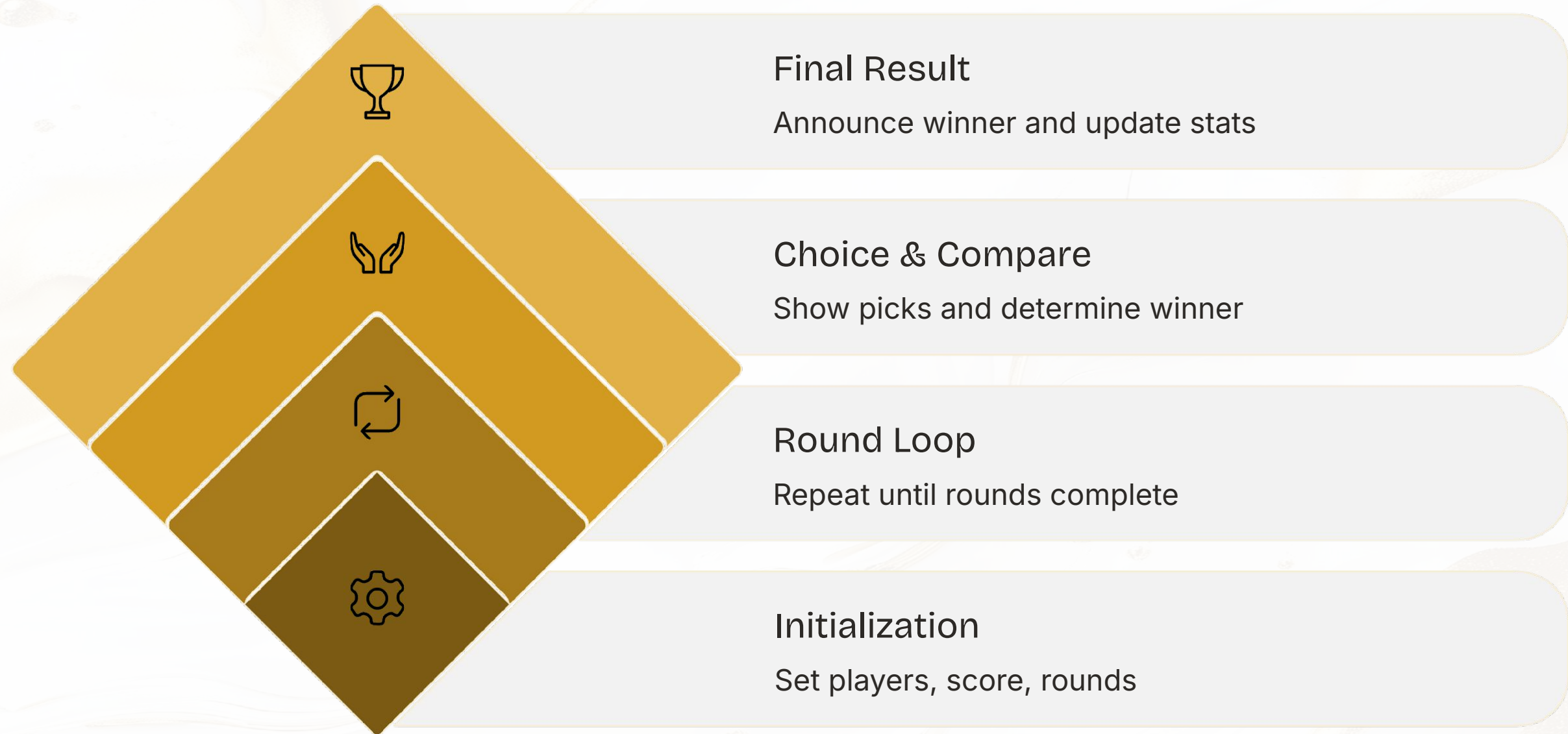
Scissors cuts Paper, decapitates Lizard

Lizard poisons Spock, eats Paper

Spock smashes Scissors, vaporizes Rock

Each gesture has specific interactions,
making the game dynamic and challenging.

Our Game's Blueprint: The Algorithm



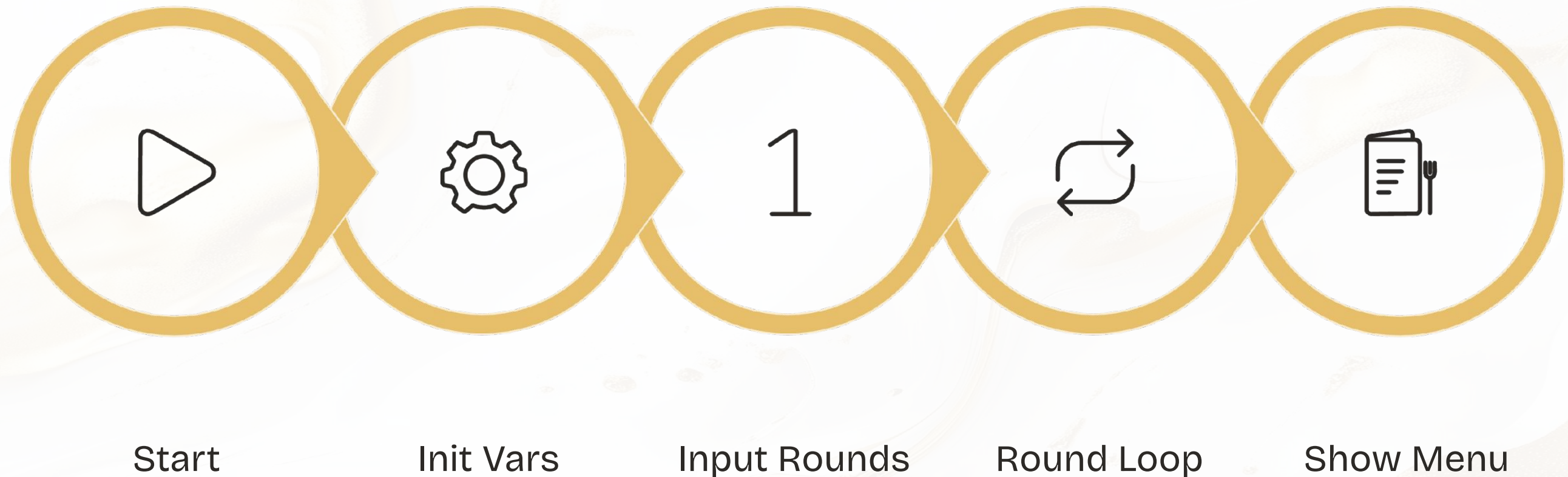
Our algorithm provides a clear, step-by-step roadmap for the game's execution, ensuring fair play and accurate scoring across multiple rounds.

Breaking Down the Logic

From setting up scores to declaring the ultimate winner, every phase is carefully planned.

01	0	0
Initialization	Round Loop	Choice Display & Comparison
Declare and set initial values for player choices (p1, p2), scores (score1, score2 = 0), and total rounds.	Iterate for the specified number of rounds. Prompt players for their choices (1-5) and capture input.	Use a <code>switch</code> statement to display choices. Compare choices: if identical, it's a draw. Otherwise, apply game rules to determine the round winner.
0	0	0
Update Scores	Final Result	
Increment the score of the winning player for the current round.	After all rounds, compare final scores and declare the overall game winner or a draw.	

Visualizing the Game Flow



This flowchart offers a visual representation of our game's operational logic, illustrating the decision points and paths of execution.

Behind the Scenes: C Code in Action

Let's look at some critical parts of the C code that bring our game to life, focusing on input, output, and game logic.

Choice Display

```
void displayChoice(int choice) {  
    switch(choice) {  
        case 1: printf("Rock"); break;  
        case 2: printf("Paper"); break;  
        // ... other cases  
        default: printf("Invalid"); break;  
    }  
}
```

This function translates numerical input into readable game choices for the players.

Main Game Loop

```
for(i = 1; i <= rounds; i++) {  
    printf("\n--- Round %d ---\n", i);  
    // Player inputs, display choices  
    // ...  
    if(p1 == p2) {  
        printf("Result: Draw!\n");  
    } else {  
        // Winning logic  
    }  
    printf("Score -> P1: %d | P2: %d\n", score1, score2);  
}
```

The core loop manages rounds, inputs, and displays the current scores.

The Heart of the Game: Winning Logic

The nested `switch` statement is where the complex rules of "Rock, Paper, Scissors, Lizard, Spock" are implemented.

```
switch(p1) {
    case 1: // Player 1 chose Rock
        if(p2 == 3 || p2 == 4) { // Rock
            crushes Scissors/Lizard
            printf("Player 1 Wins!\n");
            score1++;
        } else {
            printf("Player 2 Wins!\n");
            score2++;
        }
        break;
    case 2: // Player 1 chose Paper
        if(p2 == 1 || p2 == 5) { // Paper
            covers Rock/disproves Spock
            printf("Player 1 Wins!\n");
            score1++;
        } else {
            printf("Player 2 Wins!\n");
            score2++;
        }
        break;
    // ... cases for Scissors, Lizard,
    Spock
}
```



Intricate Decisions

This logic meticulously checks each possible combination to determine the winner of every round, embodying the true complexity and fun of the game.

It's a perfect example of using conditional statements to manage multiple outcomes.

PROJECT ACCOMPLISHMENTS

What We Achieved with This Project

This mini-project was an excellent opportunity to apply our theoretical knowledge in a practical scenario.



Functional Game

Developed a fully playable game with clear rules and scoring.



Enhanced C Skills

Deepened understanding of functions, loops, and control structures.



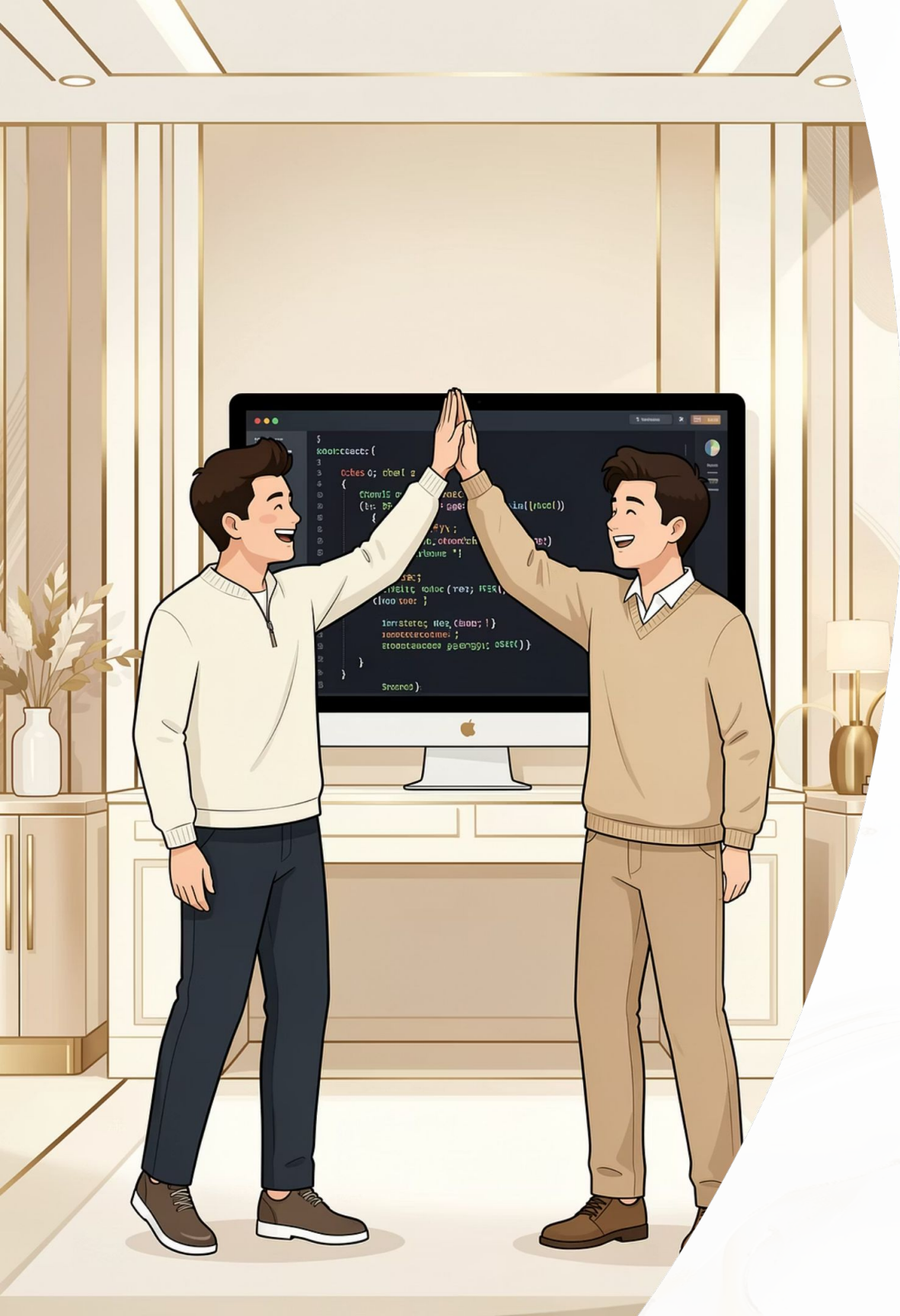
Team Collaboration

Practiced working together on a shared codebase.



Creative Problem-Solving

Applied logic to implement complex game rules effectively.



Thank You!

We hope you enjoyed our presentation on the Rock, Paper, Scissors, Lizard, Spock mini-project in C!

Any questions?

[Explore the Code](#)[Play the Game](#)